Problem:

A worker faces the following income maintenance program:

- Guaranteed income (benefit): $G = 800$ per week
- Benefit reduction rate (tax-back rate): $t = 0.4$
- Market wage: $w = 25$ per hour
- Total available time: 60 hours per week

While eligible for benefits, disposable income is given by:

$$C = G + (1 - t)wh \quad \text{for } wh \leq \frac{G}{t}$$

Once earnings exceed the break-even level, benefits are exhausted and:

$$C = wh$$

Answer the following questions clearly and show all steps.

(a) **(3 marks)** Calculate:

  (a) The break-even level of earnings.

  (b) The number of hours worked at which benefits are fully exhausted.

(b) **(2 marks)** Compute:

  (a) The effective net wage while the worker is receiving benefits.

  (b) The effective marginal tax rate on earnings.

(c) **(3 marks)** Draw the worker's budget constraint in (h, C) space. Clearly label:

  - The guaranteed income level,
  - The break-even point,
  - The slope in each region.

(d) **(3 marks)** Explain how this program affects:

  (a) The incentive to enter the labour force (extensive margin).

  (b) The incentive to increase hours worked while receiving benefits (intensive margin).

  Be precise in distinguishing income and substitution effects.

(e) **(4 marks)** Suppose the benefit reduction rate increases from $t = 0.4$ to $t = 0.7$, holding all else constant.

  (i) Explain how the budget constraint changes.

  (ii) Predict the likely effect on labour supply.

  (iii) Discuss the equity–efficiency trade-off created by increasing t.

Total: 15 marks